

Designing for Non-Engagement: How Platforms Operationalize Behavioral Silence

Syeda Ahmedi Fatima
Independent Researcher
Hyderabad, Telangana, India
fatimahmedi0803@gmail.com

Abstract

Human-computer interaction research and practice often equate successful use with visible engagement, such as posting, feedback, personalization, or consistent activity. Users who minimize such expressive interaction are commonly treated as disengaged or inactive. This paper examines how contemporary platforms respond to *behavioral silence*, focusing on system-level design choices and discourse rather than user intention. We conduct a cross-platform qualitative audit of documented platform behavior, design guidance, and user-reported experiences across social media and learning systems. Our analysis shows that silence is frequently framed as undesirable and addressed through reduced visibility, corrective nudging, or loss of accumulated progress. We derive a taxonomy of system-level silence penalties and illustrate how these mechanisms vary across platforms, including contrast cases that explicitly avoid penalizing inactivity. By grounding our analysis in documented platform practices, this work reframes non-engagement as an interactional state and highlights the risks of collapsing engagement, expression, and value in interaction design.

CCS Concepts

• Human-centered computing → HCI theory, concepts and models.

Keywords

Non-engagement, Behavioral silence, Lurking, Engagement-centric design, Platform design, Human-computer interaction

ACM Reference Format:

Syeda Ahmedi Fatima. 2026. Designing for Non-Engagement: How Platforms Operationalize Behavioral Silence. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3798568>

1 Introduction

Engagement has become a central principle in the design and evaluation of contemporary interactive systems. Across social media, learning platforms, and content-driven applications, visible engagement—such as posting, feedback, personalization, or consistent activity—is widely treated as a proxy for value. For platforms,

engagement metrics inform assessments of system health, relevance, and sustainability; for users, they increasingly shape access, visibility, and progression. As a result, many systems are explicitly designed to encourage expressive participation and consistency through algorithmic ranking, gamification, and persuasive design techniques.

Prior research shows that silence and non-participation are common and meaningful forms of interaction. Studies of online communities consistently demonstrate that most users consume content without contributing, a pattern commonly described as lurking [5]. Work on technology non-use further argues that partial use, non-use, or selective engagement can be purposeful and contextual rather than indicative of failure [7, 9]. These studies challenge deficit-based interpretations of silence but focus primarily on users' motivations and experiences.

Design responses to silence, however, are often framed as corrective. Engagement-oriented approaches such as persuasive technology and gamification aim to increase participation and consistency through incentives, nudges, and feedback loops [1, 2]. These strategies are frequently justified as necessary for system viability, particularly given participation inequality, where a small fraction of users produce most visible contributions [4]. We do not dispute the importance of engagement. However, treating expressive behavior as a proxy for engagement collapses engagement, expression, and task performance into a single behavioral signal.

This collapse has design consequences. When engagement is inferred primarily from visible expression or consistency, silence is often framed as undesirable. Platform documentation, design guidance, and interface features commonly position inactivity as something requiring intervention, whether through reduced algorithmic visibility, loss of progress indicators such as streaks, or repeated prompts encouraging re-engagement. These responses are embedded in system logic and interface design rather than explicit judgments about user intention.

Despite extensive research on lurking, non-use, calm interaction, and persuasive design [6, 8], there has been limited systematic examination of how platforms *operationalize silence* in practice. This paper addresses that gap through a cross-platform qualitative audit of documented platform behavior and discourse. Rather than explaining why users remain silent, we examine how platforms frame and respond to silence. Our contributions are threefold: (1) an empirically grounded account of how platforms treat behavioral silence, (2) a taxonomy of system-level silence penalties derived from cross-platform analysis, and (3) design implications that treat silence as an ambiguous and legitimate interactional state rather than a default failure condition.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798568>

2 Related Work

Our work engages with three primary strands of research in human-computer interaction: studies of non-use and lurking, research on calm and minimal interaction, and work on engagement-driven and persuasive design. Rather than extending any single line of work, we reposition these literatures to examine how contemporary systems respond to behavioral silence through system-level design and inference.

Non-Use, Lurking, and Passive Participation:

HCI research has long recognized that not all users actively participate in interactive systems. Early and influential studies of online communities demonstrated that the majority of users read without contributing, a phenomenon commonly referred to as lurking [5]. These studies showed that silence is widespread and often purposeful, challenging interpretations of participation inequality as mere disengagement or failure.

Beyond community participation, broader work on technology non-use has argued that not using, or partially using, systems can be meaningful and analytically valuable [7, 9]. This research re-frames non-use as contextual, situational, and socially constructed rather than as a deficit or absence. Collectively, these contributions establish silence and non-participation as common and significant interactional states.

However, this body of work primarily focuses on understanding users: their motivations, experiences, and reasons for non-participation. In contrast, our work shifts attention away from why users remain silent and toward how systems respond when silence occurs. Rather than treating silence as an outcome to be explained, we examine it as an input signal that systems operationalize through design decisions and inference.

Calm and Minimal Interaction:

Another relevant lineage is research on calm, minimal, and low-attention interaction. Weiser and Brown's vision of calm technology proposed that systems should respect users' limited attention and allow information to move fluidly between the center and periphery of awareness [8]. This work has been influential in shaping design ideals that prioritize unobtrusiveness and reduced cognitive load.

Subsequent critiques have questioned whether calmness alone adequately captures the complexities of interaction in ubiquitous computing environments. Rogers argues that an exclusive focus on calm interaction risks overlooking the active, engaged, and creative ways people use technologies, calling instead for designs that support richer forms of engagement [6]. While these perspectives highlight important tensions between calmness and engagement, they remain largely prescriptive and conceptual.

Notably, research in this area does not systematically examine how existing systems treat users who minimize interaction. While calm interaction is often framed as a desirable design goal, little attention has been paid to whether contemporary systems actually support silence, or whether they penalize it through defaults, prompts, or access constraints. Our work complements this literature by examining how silence is handled in practice rather than how it is ideally envisioned.

Engagement, Gamification, and Persuasive Design:

A substantial body of HCI research has focused on increasing user engagement through persuasive design, gamification, and behavioral incentives. Fogg's work on persuasive technology formalized how interactive systems can shape user behavior through cues, feedback, and reinforcement [2]. Building on this foundation, gamification research has examined how elements such as points, badges, and streaks motivate participation and consistency across contexts [1].

These approaches are often justified by system sustainability, learning outcomes, or community vitality, and participation inequality remains a well-documented concern [4]. However, engagement-driven design frequently treats expressive behavior as a proxy for engagement and engagement as a proxy for value. As a result, the absence of expression is commonly interpreted as disengagement, triggering nudges, restrictions, or other corrective mechanisms.

Critical perspectives have examined the ethical implications of such practices. Work on dark patterns highlights how design choices can manipulate users into behaviors they might not otherwise choose [3]. While this research raises important concerns about coercion and autonomy, it does not isolate behavioral silence as a distinct interactional condition that elicits systematic system responses.

Positioning Our Contribution:

Across these literatures, silence is alternately framed as non-use, passive participation, or a problem to be corrected through engagement-oriented design. What remains underexplored is how silence functions as a signal within system logic: how the absence of expressive interaction shapes defaults, access, prompts, and performance outcomes.

This paper addresses that gap by providing a system-centered analysis of non-engagement. Rather than explaining silence through user motivation or persuasion knowledge, we examine how systems operationalize silence through design and inference. In doing so, we position non-engagement not as a failure of users, but as a revealing case that exposes hidden assumptions in engagement-centric interaction design.

Taken together, these strands highlight a tension between recognizing silence as meaningful participation and designing systems that interpret reduced expression as disengagement, motivating a system-level examination of how silence is operationalized in practice.

3 Method

We adopt a document and discourse audit to examine how platforms respond to behavioral silence. Our analysis focuses on documented platform behavior and public discourse rather than simulated interactions or inferred user intention. The audit examined publicly available platform documentation, creator guidance materials, interface descriptions, and user discourse discussing inactivity or engagement pressures across the selected systems. Materials were reviewed iteratively to identify recurring instances where reduced activity was associated with system responses. Both penalty-oriented and neutral responses to inactivity were considered during analysis.

Platform Scope:

We examined platforms across two domains where engagement and consistency are central: (1) social media platforms, where visibility is mediated by activity signals, and (2) learning platforms, where progress is structured through consistency mechanisms such as streaks. Platforms were selected based on widespread use, explicit reliance on engagement signals, and the availability of public documentation and user discourse discussing inactivity.

Data Sources:

Our analysis draws on platform-facing documentation (e.g., creator guidelines, help pages), design and marketing discourse (e.g., advice articles), user-reported experiences from online forums, and secondary analyses from media and expert commentary. These sources are treated as complementary evidence rather than direct measures of proprietary algorithms.

Analytic Procedure:

We iteratively coded materials for instances where silence or inactivity was linked to negative outcomes such as reduced visibility, loss of progress indicators, or corrective prompting. These instances were grouped into higher-level mechanism categories, producing a taxonomy of silence penalties. We then compared patterns across platforms and identified contrast cases that explicitly avoided penalizing silence.

4 Results

Across platforms, behavioral silence is not treated as a neutral state. Instead, platforms frame inactivity as undesirable and embed corrective responses into system design and discourse.

Social Media Platforms:

On social media platforms, inactivity is closely associated with reduced visibility. Creator guidance for platforms such as Instagram and Pinterest emphasizes consistent posting to maintain reach, while extended gaps in activity are framed as leading to diminished distribution. User reports following algorithm updates frequently describe sharp drops in impressions and engagement after periods of inactivity, independent of content quality.

Learning Platforms:

Learning platforms operationalize silence through consistency mechanisms. Duolingo prominently visualizes uninterrupted activity through streaks; missed days result in visible streak loss and repeated notifications framed through emotional cues. User discourse and media analysis describe both motivational effects and negative consequences such as stress or burnout.

In contrast, Qlango explicitly markets flexible pacing and the absence of punishment for missed days. This contrast demonstrates that silence penalties are contingent design choices rather than inherent requirements of learning systems.

Unlike streak-based learning systems, inactivity in Qlango does not result in visible loss of accumulated progress or corrective notification pressure. Continued access and pacing flexibility remain unchanged despite missed activity. This contrast illustrates that engagement-supportive learning systems can operate without penalizing temporary silence.

Cross-Platform Synthesis:

Across domains, silence functions as a negative signal that triggers reduced visibility, loss of accumulated progress markers, or

corrective nudging. These responses differ in form but share a common logic: expressive consistency is treated as evidence of value, while silence results in diminished support.

Table 1: Documented system responses to behavioral silence

Platform	Silence Signal	System Response	Evidence Source
Instagram	Reduced posting	Reduced algorithmic visibility	Creator guidance; user reports
Pinterest	Inconsistent activity	Lower content distribution	Platform advice; documentation
Duolingo	Missed daily activity	Streak loss; corrective notifications	Interface features; user discourse
Qlango	Missed practice days	No penalty; flexible pacing	Platform marketing materials

Table 2: Platforms mapped to silence penalty mechanisms

Platform	Observed Mechanisms
Instagram	Algorithmic visibility loss following inactivity
Pinterest	Reduced distribution for inconsistent activity
Duolingo	Loss of progress markers; persistent nudging
Qlango	No observed silence penalties

Across platforms, these mechanisms differ in implementation but consistently translate reduced expression into signals prompting correction or reduced system support.

5 Discussion

Our findings show that platforms operationalize silence as an undesirable signal rather than an ambiguous interactional state. Engagement is necessary for many systems, but its operationalization through expressive consistency introduces structural disadvantages for quiet or intermittent users.

Silence is framed as a design problem to be corrected rather than accommodated. Platform discourse and interface features position inactivity as failure, shifting responsibility onto users while obscuring the contingency of these design choices. Contrast cases demonstrate that non-penalizing approaches to silence are viable.

We derive four design implications: (1) treat silence as an ambiguous signal, (2) avoid coupling access or visibility to expressive consistency, (3) limit moralized or emotional framing of inactivity, and (4) recognize silence as a legitimate interaction style. These implications do not reject engagement-oriented design but call for greater care in how engagement is inferred and enforced.

6 Limitations and Conclusion

This study examines a limited set of platforms using publicly available documentation and discourse. While this limits generalizability, triangulating multiple sources enables identification of recurring design patterns. We do not claim that all platforms penalize silence.

In conclusion, this paper shows that behavioral silence is treated as a negative signal across multiple contemporary platforms. By grounding our analysis in documented platform practices, we reframe non-engagement as a legitimate interactional state and highlight the risks of collapsing engagement, expression, and value in interaction design.

Author's Note on the Use of Large Language Models

In the preparation of this manuscript, the author used a large language model (ChatGPT) for language editing, refinement of academic phrasing, and to review the logical flow of arguments. This tool was applied exclusively to text developed by the author. All core ideas, theoretical framing, empirical analysis, data interpretation, and conclusions are the original work of the author.

Acknowledgments

We thank the anonymous reviewers for their valuable feedback.

References

- [1] Sebastian Deterding, Dan Dixon, Rilla Khaled, and Lennart Nacke. 2011. From Game Design Elements to Gamefulness: Defining Gamification. In *Proceedings of the 15th International Academic MindTrek Conference*. Association for Computing Machinery, New York, NY, USA, 9–15. doi:10.1145/2181037.2181040
- [2] B. J. Fogg. 2003. *Persuasive Technology: Using Computers to Change What We Think and Do*. Morgan Kaufmann, San Francisco, CA.
- [3] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (*CHI '18*). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3173574.3174108
- [4] Jakob Nielsen. 2006. Participation Inequality: Encouraging More Users to Contribute. <https://www.nngroup.com/articles/participation-inequality/>
- [5] Blair Nonnecke and Jennifer Preece. 2000. Lurker Demographics: Counting the Silent. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '00)*. Association for Computing Machinery, New York, NY, USA, 73–80. doi:10.1145/332040.332409
- [6] Yvonne Rogers. 2006. Moving on from Weiser's Vision of Calm Computing: Engaging UbiComp Experiences. In *Proceedings of the International Conference on Ubiquitous Computing*. Springer, Berlin, Heidelberg, 404–421. doi:10.1007/11853565_24
- [7] Christine Satchell and Paul Dourish. 2009. Beyond the User: Use and Non-Use in HCI. In *Proceedings of the 21st Annual Conference of the Australian Computer-Human Interaction Special Interest Group (OZCHI '09)*. Association for Computing Machinery, New York, NY, USA, 9–16. doi:10.1145/1738826.1738829
- [8] Mark Weiser and John Seely Brown. 1997. The Coming Age of Calm Technology. In *Beyond Calculation: The Next Fifty Years of Computing*. Springer, New York, NY, 75–85.
- [9] Sally Wyatt. 2003. Non-Users Also Matter: The Construction of Users and Non-Users of the Internet. In *How Users Matter: The Co-Construction of Users and Technology*. MIT Press, Cambridge, MA.